Scope change: existing annotations instead of new hand labeling

Pragmatic Intent Detection for AI Assistants · Sandy Thomas · Machine & Deep Learning, Fall 2026 · October 13, 2026

Summary. I am replacing the planned hand labeling of 160–240 test turns (three annotation passes each, roughly 10–13 hours) with annotations that already exist: DIRECT's human-written direct and indirect rewrites of each MultiWOZ user turn, MultiWOZ's dialogue-act labels, and fixed wording rules that assign each utterance its indirectness level. The research question, the two-stage task, the context conditions and the model panel from my proposal do not change. I would like your OK on this change.

1. What changes and what stays

Proposal element	Plan as of Oct 12	Now
Research question, two-stage task (detect, then interpret)	As proposed	Unchanged
Context conditions	None / one prior turn / full history / mismatched control	Unchanged
Model panel	TF-IDF + LR, RoBERTa/DeBERTa, an LLM, a two-stage hybrid	Unchanged
Indirectness levels (L0–L3)	Hand-labeled on 160–240 test turns	Assigned by fixed wording rules to every test utterance, checked against my hand labels
Intents and Stage 2 gold	Hand-labeled intent, slots and paraphrase	MultiWOZ dialogue acts, mapped to four of the proposal's intent categories
Test set	160–240 newly labeled turns	DIRECT's test split (MultiWOZ's official test dialogues)
Human time still needed	About 10–13 hours of labeling	About 1 hour of short checks

2. How the levels are assigned

The levels follow the CCSARP request-strategy scale (Blum-Kulka, House & Kasper, 1989) as written in my annotation rubric (v2). The rules read the wording only and apply in order; the first that matches wins, so the most direct request in an utterance sets its level. Examples are invented, taken from the rubric.

Level	Rule	Example
L0 direct	Command, need/want statement, or plain question	“Book a table for two.” “I need a train after 6pm.”
L1 conventionally indirect	Stock formula: can/could/would you, can I get, is it possible, how about	“Could you book a table for two?”
L2 strong hint	No formula, but names a detail of what is wanted (day, time, place, party size, item)	“There are two of us, and we're free around seven.”
L3 mild hint	No formula and names nothing that is wanted	“It's our anniversary tonight and I haven't planned a thing.”
INF / NR	Gives information only / thanks, goodbye or declines	“No preference on the area.”

3. Do the rules agree with a human?

I wrote the rules from the rubric's text and invented examples only, froze them, and then scored them once against the 69 utterances I had already labeled by hand in three calibration rounds.

Comparison	Utterances	Agreement	Cohen's κ
Rounds 1 and 3 (already adjudicated)	39	79%	0.74
Round 2 (not yet adjudicated; reported separately)	30	43%	0.28

For reference, my plan set $\kappa \geq 0.6$ as the bar for agreement between two human annotators. Round 2's labels were made before the rubric was settled and have not been adjudicated, so they are not a fair test yet; after adjudication they become a fresh check. Seven of the eight misses on rounds 1 and 3 fall into two patterns: plain answers with no detail labeled L3 instead of INF, and "I need to leave at …" labeled L0 instead of L2, a case where the rubric itself is inconsistent and I need to settle it.

4. Coverage, and a finding about DIRECT

The rules level every utterance, so the test split has far more than the 25–40 examples per level the plan aimed for. Counts for DIRECT's test-split rewrites (turns with dialogue acts):

DIRECT rewrite	L0	L1	L2	L3	INF	NR
"Direct" rewrites	3,349	985	970	79	132	1,186
"Indirect" rewrites	1,837	1,375	1,784	352	95	1,258

By wording, about 27% of DIRECT's "indirect" rewrites are direct requests (L0), and about 15% of its "direct" rewrites use a conventional formula (L1). DIRECT's two labels alone are therefore a noisy measure of indirectness. Using the graded levels addresses this and is itself a result worth reporting.

5. Limitations I will state in the paper

- The levels are rule-assigned, not human gold. Every result that uses them reports the agreement above.
- Strong and mild hints (L2/L3) have the least human evidence behind them. A 40-utterance spot check of the rules' L3 is scheduled.
- The intents cover four of the proposal's categories (information seeking, booking/action requests, travel planning and navigation, recommendation). Troubleshooting, advice and device control are not in this data.
- There is one annotator. A second person labeling about 50 utterances would make the agreement check independent of me.
- DIRECT's authors already tested adding dialogue history. This study's contribution is the graded context conditions, the mismatched-context control, the model-family comparison and the level-by-context interaction.

6. Questions for you

- Is this change acceptable for the class paper?
- Optionally, could someone independently label about 50 utterances so the agreement check doesn't rest on my labels alone?